Tracking AI Trends Through Natural Language Processing (NLP)-Powered Text Mining and Knowledge Extraction

From Exploratory Text Analysis to Large Language Model (LLM) Applications

Background

In today's fast-changing AI landscape, timely and accurate information is crucial. AI breakthroughs in generative models, robotics, natural language processing (NLP), computing infrastructure, and emerging agentic systems are reshaping industries and society. This project equips students with computational language technologies to analyze and interpret these developments.

The project uses NLP techniques to analyze media discourse, uncover structured knowledge, and track innovation in AI. Students develop practical experience with text analytics, representation learning, knowledge graphs, and retrieval-augmented generation (RAG).

The AI Media Dataset (September 2024 - August 2026) serves as the foundation for the project. The extended corpus of curated media texts offers material to:

- study AI innovations and debates,
- track emerging trends over time within a selected technology domain,
- analyze sentiment and discourse around key developments,
- build knowledge graphs and intelligent Q&A systems for AI trend monitoring.

By the end of the project, students should be able to design and evaluate an end-to-end NLP workflow for trend detection and knowledge extraction, and critically assess how different methods support reliable analysis of fast-moving AI developments.

Goals

The project has three main goals:

- **Advanced Text Analytics Mastery.** Expand skills in NLP techniques, from topic modeling to embeddings and LLMs, to extract insights from media texts on AI.
- **Trend Analysis and Innovation Mapping.** Track developments in a selected AI technology domain, identify emerging topics, and analyze how discourse evolves over time.
- **Applied Knowledge Systems.** Build Q&A and information retrieval systems that integrate text retrieval, knowledge graphs, and RAG for trend monitoring and decision support.

Organizers

The project is organized by:

Dr. Janna Lipenkova, CEO, Anacode GmbH, Germany
Dr. Guang Lu, Lecturer for Data Science, Hochschule Luzern (HSLU), Switzerland

in collaboration with:

Dr. Diego Antognini, Senior Research Scientist, Google DeepMind, Switzerland

as a joint industry-academia project for the HSLU Applied Data Science and AI Master's Program in the course Computational Language Technologies.

Project Pipeline Overview

The capstone project consists of three interconnected stages. Each group **selects one topic track at the beginning of the project** and uses it as the thematic focus for knowledge graph construction and downstream RAG tasks.

- **Stage 1: Data Cleaning, Preprocessing, and Exploratory Analysis.** Explore and clean the dataset, analyze trends, extract entities and relations, and construct a topic-focused knowledge graph.
- **Stage 2: Advanced Embedding Models Training and Analysis.** Train classical embeddings and fine-tune compact open models, compare their performance, and use the selected topic as a focused analysis case.

- **Stage 3: Implementing an RAG System for Question Answering.** Build and evaluate a RAG system that combines text retrieval with structured knowledge from the selected topic track.

Together, the three stages cover the end-to-end lifecycle of an NLP project: raw text data -> preprocessing and knowledge extraction -> embeddings and fine-tuning -> RAG-based question answering with text and graph evidence.

Task

We provide one comprehensive dataset:

The AI Media Dataset: https://www.kaggle.com/datasets/jannalipenkova/ai-media-dataset

Students will explore this dataset through the three-stage pipeline. The pipeline is an important reference for the project work and serves as the basis for assessment.

Topic Tracks - Select One

Each group selects **one** of the following three tracks at the beginning of the project. The selected track should be used consistently for the Stage 1 knowledge graph and the Stage 3 Q&A/RAG tasks. Stage 2 may use the full corpus, but should include at least one analysis focused on the selected track.

1. AI governance

Developments related to AI security, accountability, bias and fairness, and the regulatory landscape.

2. Open models

Monitor the emergence and development of open-source/open-weight models, including their core characteristics, architectures, capabilities, developers, model families, and relationships to earlier models.

For orientation, students may use the AI Radar overview of model families: AI Radar - Model Families

3. Agentic Web

Technologies, standards, tools, and developments related to autonomous AI agents and their interaction with web-based systems, services, data, and other agents.

For orientation: AI Radar - Agentic Web

AI Radar is intended as an orientation resource for scoping concepts and terminology. Evidence used in the project knowledge graph and RAG evaluation should remain traceable to the AI Media Dataset unless explicitly documented otherwise.

Stage 1: Enhanced Data Cleaning, Preprocessing, and Exploratory Analysis

Objective

Explore and preprocess the AI Media Dataset to uncover emerging trends, entities, and relationships within the selected topic track. The main outputs are an analytical overview and a topic-focused knowledge graph that can later be used in Stage 3.

Data Collection and Cleaning

- Download and load the AI Media Dataset.
- Clean the data: remove duplicates, handle missing values, and normalize text (e.g., casing, HTML, irrelevant symbols).

Text Preprocessing

- Apply tokenization, lemmatization, and stop-word removal where appropriate.
- Ensure consistent representation across documents, including acronyms and abbreviations relevant to the selected track.

Exploratory Data Analysis (EDA)

- Conduct temporal analysis over September 2024 - August 2026, with a focus on how the selected topic develops over time.

- Perform sentiment or stance analysis where meaningful for the selected topic.
- Generate clear visualizations such as bar charts, time-series plots, or term/topic trajectories.

Entity & Relation Extraction

- Use pre-trained NER models (e.g., spaCy, Hugging Face) to identify relevant companies, organizations, technologies, models, standards, products, and people.
- Apply LangExtract (GitHub) or a comparable structured extraction approach to improve entity and relation extraction.
- Build a co-occurrence matrix showing which entities frequently appear together.

Knowledge Graph Construction

- Define a compact topic-specific schema before graph construction, including the main entity and relation types.
- Construct a graph in which nodes represent entities and edges represent co-mentions or extracted relations.
- Analyze the graph using network metrics (e.g., centrality, clustering) to reveal influential entities and emerging relationships.
- Visualize the graph with tools such as NetworkX or Graphviz.
- Store the graph and its schema so both can be reused in Stage 3.

Topic Modeling

- Apply models such as LDA, NMF, Top2Vec, or BERTopic to uncover major themes.
- Use hierarchical or dynamic topic analysis where useful to reveal subtopics and their evolution.
- Interpret topic distributions over time and relate them to the selected track.

Outputs

- Notebook with data cleaning and preprocessing steps.
- Visualizations of temporal trends, sentiment/stance where relevant, and topic modeling results.
- A topic-focused knowledge graph plus a short schema/data dictionary.
- Documentation of key insights and notable developments in the selected track.

Stage 2: Advanced Embedding Models Training and Analysis

Objective

Develop and evaluate embedding and language models tailored to the AI Media Dataset. Students explore classical representations and compact open models, and use the selected topic track as a focused case for interpreting model behavior and trend signals.

Data Preparation for Embeddings

- Preprocess the AI Media Dataset for training and prepare suitable training/validation splits.

Classical Word & Sentence Embeddings

- Train embeddings using Word2Vec, FastText, or Doc2Vec.
- Visualize semantic clusters and thematic overlaps in AI discourse, including terms relevant to the selected track.

Fine-Tuning Compact Open Models

- Select one or more compact open models suitable for Colab Pro or similar environments. Candidate models may include:
  - Qwen3-1.7B / Qwen3-4B
  - Gemma 4 E2B / E4B
  - Ministral 3 3B
  - SmolLM3-3B
  - Apertus-v1.1-1.5B / Apertus-v1.1-4B
  - Apertus-v1.5-8B (where computational resources permit)
- Use parameter-efficient fine-tuning such as LoRA/QLoRA rather than implying full fine-tuning of these models.

Comparative Evaluation

- Evaluate embeddings and fine-tuned models using intrinsic tasks such as semantic similarity, clustering quality, and visualization.
- Use at least one extrinsic task such as trend/topic classification, sentiment/stance detection, or another justified task relevant to the selected track.
- Compare trade-offs in model size, performance, training speed, and GPU requirements.

Insight Extraction

- Use the best-performing models to analyze AI discourse in the dataset.
- Include a focused analysis of the selected track and explain which representations capture its developments most effectively.

Outputs

- Notebook with model training and fine-tuning steps, clearly annotated.
- Visualizations comparing embeddings and model outputs.
- Short summary of model trade-offs and insights from the selected topic track.

Stage 3: Implementing an RAG System for Question Answering

Objective

Build and evaluate a RAG system over the AI Media Dataset for the selected topic track. The system should integrate text retrieval and a knowledge graph to support accurate, context-grounded, and time-aware question answering about AI developments.

Recommended resources:

- Toolbox with RAG tools: https://github.com/umagira/anatoolbox/tree/main
- Baseline implementation: https://github.com/umagira/anatoolbox/blob/main/notebooks/course_project_skeleton.ipynb
- Dr. Janna Lipenkova, *The Art of AI Product Development*, Chapter 7 on RAG systems. Google Docs

Q&A Dataset Construction

- Select approximately 100–150 representative passages from the selected topic track, covering different subtopics, entities, and time periods.
- Use an LLM that is different from the model used in the RAG pipeline to generate approximately 200–300 Q&A pairs, typically 1–3 questions per passage.
- Include a mix of factual, temporal, relational, analytical, and comparative questions.
- Manually review and refine the Q&A pairs for clarity, correctness, diversity, coverage, and answerability from the source evidence.

RAG Implementation

- Refer to an existing RAG implementation (as in this notebook) to set up a text-based RAG baseline.
- Experiment with retrieval strategies such as dense retrieval, sparse retrieval, or hybrid search.
- Ensure answers are grounded in retrieved passages and include publication dates when relevant to temporal questions.

Knowledge Graph Integration

- Integrate the group's Stage 1 knowledge graph for the selected topic into the RAG system.
- Explore graph-aware or entity-anchored retrieval strategies that combine structured graph evidence with unstructured text retrieval.
- Compare the graph-enhanced RAG system with the text-only baseline to assess its contribution to retrieval quality, answer quality, and temporal/trend-oriented questions.
- Ensure answers reference relevant graph entities and relations where appropriate.

System Evaluation

- Evaluate Q&A performance across categories using metrics such as:
  - Faithfulness - accuracy against source evidence
  - Relevance - directly addressing the query

    - Context grounding - appropriate use of retrieved passages and graph evidence
    - Temporal grounding - correct use of dates and chronology where relevant
- Compare performance of:
    - Text-only RAG
    - RAG with the group's Stage 1 knowledge graph

RAG Enhancement

- Identify one limitation of the baseline or graph-enhanced RAG system based on the evaluation results.
- Select and implement one feasible enhancement, such as re-ranking, query expansion, graph-based retrieval expansion, improved chunking, or retrieval filtering.
- Evaluate whether the enhancement improves retrieval or answer quality.
- Briefly discuss the observed benefits, limitations, and trade-offs.

Outputs

- Notebook with the full RAG pipeline and graph integration.
- Evaluation tables/visualizations comparing text-only and graph-augmented approaches.
- Implementation and evaluation of one RAG enhancement, with a discussion of its impact and limitations.
- Final insights into how RAG + knowledge graphs support trend monitoring in the selected topic track.

Requirements

(1) Students form groups of three to tackle the project as a team.

(2) Each group selects one of the three topic tracks at the beginning of the project and keeps this focus for Stage 1 knowledge graph construction and Stage 3 RAG tasks.

(3) A separate final report is not required. However, clear code structure, annotations, explanations, and visualization of results are expected. The notebook should ideally read as a reproducible tutorial.

(4) Each group member should clearly indicate their contribution in the notebook.

(5) The project can be started with Google Colab. If additional compute or RAM is required, students may apply for Colab Pro or Colab Pro+; reimbursement arrangements will be communicated by the study program.

(6) The use of AI coding tools is permitted and encouraged, but must be disclosed in the submission, including the tool name and how it was used. Non-disclosure constitutes a violation of the project requirements.

(7) Submit the notebook for each of the three stages according to the deadlines and submission guidance communicated on ILIAS.